A Request For Proposal on

Development of Student Verification and Transcript API System for Far Western University

Introduction:

The rapid digitization of public services has created a strong need for secure, real-time, and interoperable academic data systems. Far Western University currently maintains student academic records within its internal examination system; however, these records are not easily accessible for external verification or integration with national digital platforms.

The Government of Nepal's Nagarik App aims to centralize citizen services, including academic verification. To support this initiative, Far Western University proposes the development of a **Student Verification and Transcript API System** that enables secure integration with the Nagarik App.

This system will allow students, institutions, and authorities to access verified academic records digitally, improving efficiency, transparency, and service delivery.

The existing system faces several limitations:

- **Manual Verification Process**: Academic verification requests are handled manually, causing delays.
- **Lack of Integration**: No direct connection between the university system and external platforms like Nagarik App.
- **Time-Consuming Transcript Requests**: Students must physically visit or submit manual forms.
- **Limited Accessibility**: Students outside the region face difficulty accessing services.
- **Security Risks**: Lack of standardized APIs for secure data exchange.

With the Government of Nepal mandating online verification systems through the Nagarik App initiative, FWU needs to:

- Enable instant verification of student enrollment
- Provide programmatic access to academic records
- Comply with national digital transformation standards
- Reduce administrative burden on university staff

Proposed Solution:

The **API-based system** should have the following capabilities:

- **Push API**
    - **Students Verification API:** Fetch student enrollment detail for the verification from the FWU system into Nagarik App.
    - **Transcript/Certificate API**: Fetch student transcript/certificate records from the FWU system into Nagarik App
- **Pull API**

    - Fetch student transcript/certificate request form data from Nagarik App.

- **Admin Dashboard at FWU Exam System**
    - **Transcript/Certificate Request View**
    - **Audit Logging**:
        - Automatic logging of all API requests
        - Records: timestamp, registration number, request outcome, IP address
        - Supports compliance and accountability requirements

Scope of the Work:

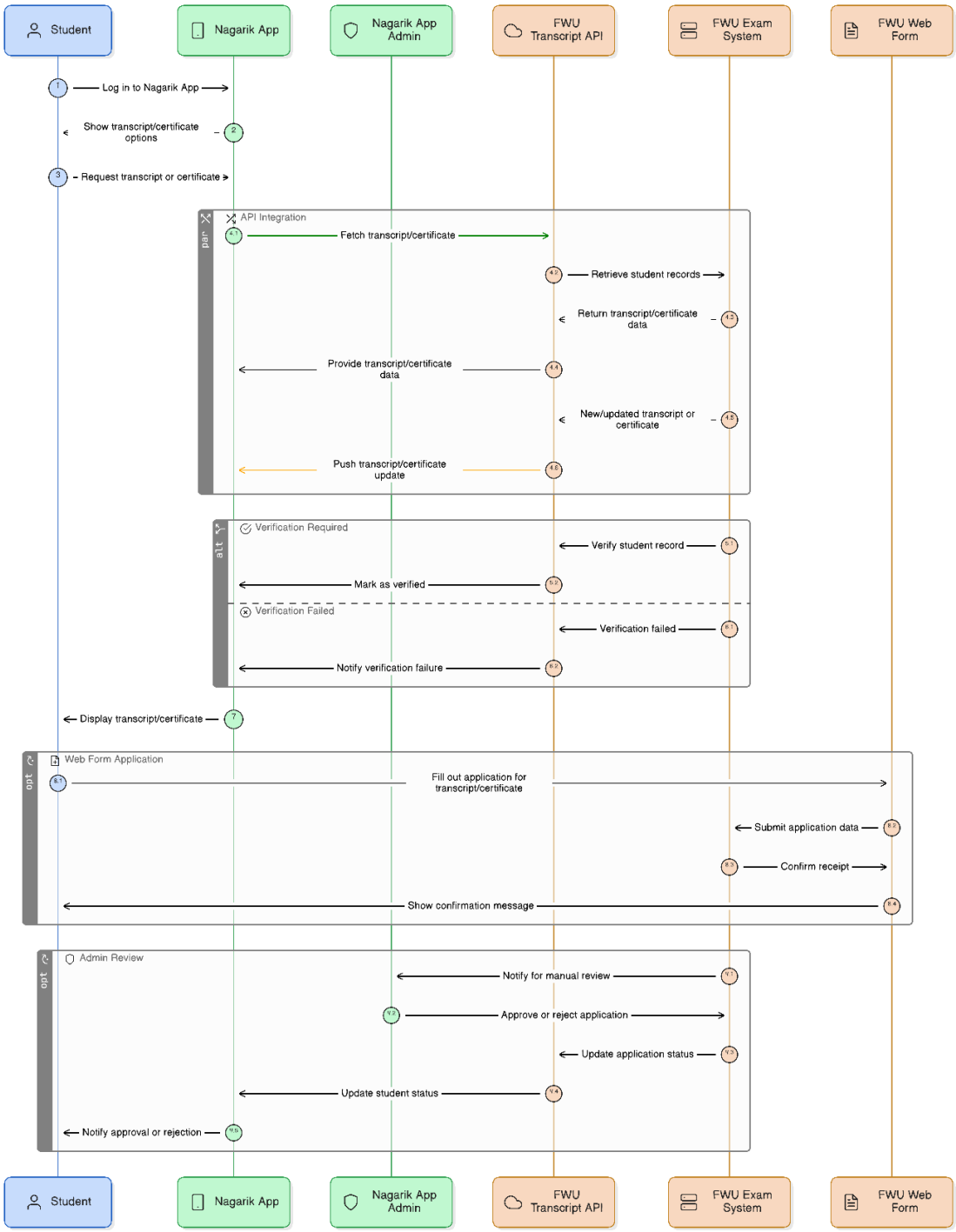
The scope of this project is as below;

Phases	**Deliverables**
Requirement Analysis	SRS Document, API Specification, Database Schema
API Development	Student Verification Endpoint, Transcript/Certificate Endpoint, Error Handling
Database Setup	SQL Server Database, Entity Models, Migrations, Indexes
Security Implementation	Input Validation, SQL Injection Prevention, Rate Limiting
Testing	Unit Tests, Integration Tests, User Acceptance Testing
Documentation	API Documentation, User Guide, Deployment Guide
Deployment and Integration	Cloud Infrastructure Setup, SSL Configuration, Integration with FWU Exam System and Nagarik App
Support (1 Year)	Bug Fixes, Security Updates, Performance Optimization

Technology Stack:

- API Framework: ASP.NET Core
- Minimal API Style
- ORM Entity Framework Core
- SQL Server Database
- Appropriate Cloud Platform

System Flow Architecture:



Database Schema:

The Database schema will be as per the FWU Exam System.

Security Measures:

The system should incorporate a comprehensive set of security measures to ensure data protection, integrity, and safe access.

- All user inputs should be strictly validated and sanitized to prevent malformed or malicious data from entering the system.
- SQL injection attacks should be mitigated through the use of parameterized queries and secure ORM practices.
- Communication between systems should be enforced over HTTPS with TLS encryption to ensure secure data transmission.
- Authentication should be handled using API keys and/or JWT tokens to verify and authorize legitimate users and services.
- Rate limiting mechanisms should be implemented to prevent abuse, denial-of-service attempts, and excessive API usage.
- In addition, continuous security monitoring and logging should be maintained to detect suspicious activities, ensure compliance, and support timely incident response.

Performance Considerations:

To improve the performance and scalability of the Student Verification and Transcript API System, several optimization techniques should be implemented. They should feature following;

- Database indexes on key attributes such as *RegdNo* and *DobAD* ensure faster data retrieval during student verification queries.
- Connection pooling to reuse existing database connections, reducing overhead and improving system efficiency under heavy load. A
- Synchronous programming using async/await enables the system to handle multiple requests concurrently without blocking operations, resulting in better responsiveness.
- Response caching should be applied for frequently accessed data to minimize repeated database calls and significantly reduce latency.
- Additionally, pagination should be used for large datasets to deliver data in manageable chunks, improving both API performance and user interface experience.

Together, these techniques ensure a fast, scalable, and efficient system suitable for integration with the Nagarik App.

Implementation Timeline:

Phases	Duration	Deliverables
1. Requirement Analysis	1 week	SRS, API Spec, DB Schema
2. Database Setup	1 week	DB Creation, Entities, Seeds
3. API Development	2 weeks	Endpoints, Business Logic
4. Security & Best Practices	1 week	Auth, Validation, Logging
5. Testing	1 weeks	Unit Tests, Integration Tests, UAT
6. Deployment and Integration	1 weeks	Production Deployment
Total	**7 Weeks**	

Estimated Budget: (For Office Consent Only):

Description	Amount	Remarks
API Design, Development, Deployment and Integration	5,00,000/- (Five Lakhs Only)	One Time Cost
Infrastructure Development (Cloud Service Hosting, DB Platforms, SSL Service, Backup and Monitoring	60, 000/- (Sixty Thousands Only)	Annual Cost

* The above charges are exclusive of annual maintenance charges.

Annual Maintenance Service:

An annual maintenance service is require to maintain smooth execution of the service. An agreement including terms and condition along with annual maintenance charges should be done between the FWU and Service Provider.